

Algebra Practice

Expand

a) Simplify $a(5b - a) - (b - a)^2$.

b) Factorise $6d^2 + 30de - 10e - 2d$ completely.

a) Simplify $5(3x - 5)^2 - 3(5 - 3x)$.

b) Factorise completely $-25y^2 - 5xy + x + 1$.

- (a) Expand and simplify $(2x - 1)(2 - 3x) - 3x(2x - 5)$.
(b) Factorise completely $24ab - 4ac + pc - 6pb$.

(a) Factorise completely $16p^4 - 81q^4$.

(b) Factorise $9x^2 - 12xy + 4y^2$.

Make t the subject in the formula $x = 2t \sqrt{\frac{k^2}{2k^2 + 3t^2}}$.

$$W = 6 \sqrt{\frac{y^3 - x^2}{7}}$$

(a) Calculate the value of W when $y = 5$ and $x = -4$. Write your answer correct to two decimal places.

(b) Rearrange the formula to make x the subject.

It is given that $1 - \frac{a-b}{b+2c} = \frac{2a-1}{2}$.

Express b in terms of a and c .

Write as a single fraction in its simplest form $\frac{5x}{(3-x)^2} - \frac{1}{x-3}$

Express $\frac{3}{2-3x} - \frac{24x}{9x^2-4} + \frac{4}{3x+2}$ as a single fraction in its simplest form.

(a) Express as a single fraction $\frac{7}{(6-5p)^2} - \frac{2p-1}{10p-12}$.

(b) Simplify $\frac{6x^2-17x+5}{18x^2-2} \times \frac{15x+5}{10-4x}$.

Simplify $\frac{4x^3-25xy^2}{2x^2+7xy-30y^2}$.

Fraction multiplication & division. ☆☆

Steps:

- ① If there is $\div$, do keep, change and flip

$$\text{e.g. } \frac{5x+10xy}{2x+6} \div \frac{2x+4xy}{7x+21}$$

$$\frac{5x+10xy}{2x+6} \times \frac{7x+21}{2x+4xy}$$

- ② If there is multiple variables, factorise first $\Rightarrow$ Types of factorisation

$$\text{e.g. } \frac{5x+10xy}{2x+6} \times \frac{7x+21}{2x+4xy}$$

$$\frac{5x(1+2y)}{2(x+3)} \times \frac{7(x+3)}{2x(1+2y)}$$

- ① common factors

$$\text{e.g. } 5x+10y = 5(x+2y)$$

- ② quadratic eqn (use calculator!)

$$\text{e.g. } x^2+x-6 = (x+3)(x-2)$$

- ③ cancel common factors by looking diagonally

$$\frac{5\cancel{x}(1+2y)}{2\cancel{(x+3)}} \times \frac{7\cancel{(x+3)}}{2x(1+2y)}$$

$$= \frac{5}{2} \times \frac{7}{2}$$

$$= \frac{35}{4}$$

- ③ $a^2-b^2 = (a+b)(a-b)$

$$\text{e.g. } 9x^2-25$$

$$= (3x+5)(3x-5)$$

Common mistakes:

① $\frac{4x^2}{5} \times \frac{1}{x}$ ✓

$$= \frac{4x}{5} \times \frac{1}{1}$$

$$= \frac{4x}{5}$$

$\frac{4x^2}{5} \times \frac{1}{x}$ ✗

$$= \frac{4x^2}{5} \times \frac{1}{x}$$

$$= \frac{16}{5}$$

② $\frac{8x+16y}{4} \times \frac{5}{7x+14y}$ ✓

$$= \frac{8\cancel{(x+2y)}}{4} \times \frac{5}{7\cancel{(x+2y)}}$$

still can simplify

$$= \frac{8^2}{4} \times \frac{5}{7}$$

$$= \frac{2}{1} \times \frac{5}{7} = \frac{5}{7}$$

$\frac{8x+16y}{4} \times \frac{5}{7x+14y}$ ✗

$$= \frac{8\cancel{(x+2y)}}{4} \times \frac{5}{7\cancel{(x+2y)}}$$

$$= \frac{8}{4} \times \frac{5}{7}$$

$$= \frac{40}{28} \rightarrow \text{still can simplify}$$

Simplify

(a) $\frac{2a^2}{5bc} \div \frac{14a}{15c}$

(b) $\frac{36}{2p-14q} \times \frac{p^2-49q^2}{3}$

(b) Simplify $\frac{y^2+2y-3}{2y-10} \div \frac{(y-1)^2}{y-5}$.

Simplify the following algebraic fractions.

(a) $\frac{b}{7c^2} \times \frac{2}{bc} \div \frac{1}{c^2}$.

(b) $\frac{4x^2-y^2}{2x+y} \times \frac{5}{x^2-6x-7}$

Simplify $\frac{x}{x^2-4} - \frac{2}{2-x}$.

Solving Quadratic Eqns.

A quadratic equation is of the form

$$ax^2 + bx + c = 0$$

where a, b and c are real numbers and $a \neq 0$.

Solve the equation $x^2 + 6x + 8 = 0$.

Solve the following equations, giving your answers to 3 significant figures where necessary.

(i) $(x+2)^2 = 25$

(ii) $(x-5)^2 = 20$

QUESTION 1

Solve the following equations, giving your answers to 3 significant figures where necessary.

(i) $(3x-2)^2 = 16$

(ii) $(5-x)^2 = 7$

(iii) $\left(\frac{1}{2}x-4\right)^2 = 81$

WHERE PASSIONATE TEACHING INSPIRES
www.timganmat.com

(iv) $(2x+5)^2 = 45$

(v) $(x-5)^2 = \frac{9}{16}$

(vi) $\left(2x+\frac{3}{2}\right)^2 = \frac{81}{100}$

Solve the following quadratic equations, giving your answers correct to 2 decimal places.

(i) $x^2 + 6x - 5 = 0$

(ii) $x^2 - 5x + 3 = 0$

Solve the following equations by completing the square, giving your answers correct to 2 decimal places.

(i) $x^2 + 3x + 6 = 0$

(ii) $x^2 - 5x - 3 = 0$

(iii) $5x - x^2 + 4 = 0$

(a) Solve the equation $\frac{1}{(x+1)(x-2)} = \frac{2}{8-x}$.

Solve the equation $\frac{2}{x-4} + \frac{7x}{3x-2} = 1$.

Use the quadratic formula to solve the equation.

$$3x^2 - 7x - 4 = 0$$